🎓 Education Regulation Impact Analyzer (ERIA)

📌 Project Overview

Education Regulation Impact Analyzer (ERIA) is an AI-powered NLP and Large Language Model (LLM) application designed to simplify complex education policies, regulations, circulars, and notifications issued by educational authorities such as UGC, AICTE, NAAC, NIRF, and universities.

The system automatically analyzes uploaded regulation documents and generates easy-to-understand summaries, stakeholder impact reports, risk assessments, and policy insights.

🚀 Features

📄 PDF Regulation Analysis

- Upload education regulation documents in PDF format
- Automatic text extraction and preprocessing

🏷️ Regulation Topic Classification

Automatically categorizes regulations into:

- Accreditation
- Scholarship
- Admissions
- Curriculum
- Faculty Policy
- Examination
- Research Policy
- University Governance

🧠 AI-Powered Summarization

Generates:

- Executive Summary
- Purpose of Regulation
- Key Changes
- Opportunities
- Recommendations

👥 Stakeholder Impact Analysis

Identifies impact on:

- Students
- Faculty
- Universities

- Colleges
- Administrators
- Accreditation Teams

⚠️ Risk Assessment

Detects:

- Compliance Challenges
- Administrative Burden
- Academic Risks
- Institutional Readiness Issues

📈 Impact Forecasting

Provides:

- Short-Term Impact (0–1 Year)
- Medium-Term Impact (1–5 Years)
- Long-Term Impact (5+ Years)

🎨 Interactive Dashboard

Built using Gradio for:

- PDF Upload
- AI Summary Generation
- Topic Detection
- Impact Analysis
- Risk Assessment

🏗️ Project Architecture

```
PDF Upload
     |
     ▼
PDF Text Extraction
     |
     ▼
Text Cleaning & NLP Processing
     |
     ▼
Topic Classification
     |
     ▼
Keyword Extraction
     |
     ▼
Stakeholder Detection
```

```
      │
      ▼
Gemini LLM Analysis
      │
      ├── Summary Generation
      ├── Impact Analysis
      ├── Risk Detection
      └── Recommendations
      │
      ▼
Gradio Dashboard
```

🛠️ Tech Stack

Category	Technology
Programming Language	Python
Development Environment	Google Colab
LLM	Gemini 1.5 Flash
NLP	NLTK, SpaCy
Topic Classification	Hugging Face Transformers
Keyword Extraction	KeyBERT
PDF Processing	PyMuPDF
Dashboard	Gradio
Visualization	Plotly
Deployment	Hugging Face Spaces

📂 Project Structure

```
ERIA/
│
├── data/
│   ├── raw_pdfs/
│   └── processed/
│
├── notebooks/
│   └── ERIA_Project.ipynb
│
├── src/
│   ├── pdf_parser.py
```

```
│   ├── preprocessing.py
│   ├── classifier.py
│   ├── impact_analyzer.py
│   ├── summarizer.py
│   └── report_generator.py
│
├── app.py
├── requirements.txt
├── README.md
└── presentation.pptx
```

⚙️ Installation

Clone the repository:

```
git clone https://github.com/yourusername/ERIA.git
cd ERIA
```

Install dependencies:

```
pip install -r requirements.txt
```

Download SpaCy model:

```
python -m spacy download en_core_web_sm
```

🔑 Gemini API Setup

Get your API key from:

https://aistudio.google.com/

Add your API key:

```
GEMINI_API_KEY = "YOUR_API_KEY"
```

▶️ Running the Project

Google Colab

1. Open `ERIA_Project.ipynb`
2. Install dependencies
3. Add Gemini API key
4. Upload regulation PDF
5. Run all cells

Gradio Dashboard

```
interface.launch()
```

📊 Sample Output

Detected Topic

```
Scholarship Policy
```

Executive Summary

```
The regulation introduces revised scholarship
guidelines aimed at increasing accessibility
for economically weaker students.
```

Stakeholder Impact

```
Students → High Positive Impact

Faculty → Low Impact

Institutions → Moderate Compliance Changes
```

Risk Assessment

```
Additional documentation requirements

Potential implementation delays

Compliance monitoring burden
```

📈 Evaluation Metrics

- Topic Classification Accuracy
- Summary Quality
- Stakeholder Detection Accuracy
- Risk Detection Relevance
- User Readability Score
- Processing Time

🔮 Future Enhancements

- Retrieval-Augmented Generation (RAG)
- FAISS Vector Database
- Policy Similarity Search
- Chat with Regulation Documents
- Knowledge Graph Generation
- Multi-Document Comparison
- PDF Report Export
- LangFlow Integration

🎯 Project Deliverables

- Google Colab Notebook
- Gradio Dashboard
- GitHub Repository
- Project Documentation
- PPT Presentation
- Demo Video
- Hugging Face Deployment

👨‍💻 Author

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AI/ML | NLP | LLM Projects

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